

Occurrence data-Model strategy



openModeller

<http://openmodeller.sourceforge.net>

Input-occurrence data

- Latitude/longitude

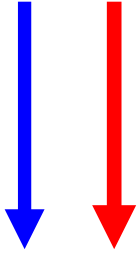
- Format:

- Degree/minute/second

- Decimal format

Location/site → coordinates (lat/long)

Degree, minute, second

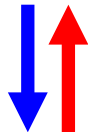


Decimal degrees

Pullman:

Latitude: 46° 43' 57.36" N

Longitude: 117° 10' 18.48" W



Latitude: 46.7298°

Longitude: -117.1817°

Prime Meridian

0° longitude



North
America

Europe

Latitude: positive coordinates
Longitude: negative coordinates

Latitude: positive coordinates
Longitude: positive coordinates

Equator
0° latitude

Africa

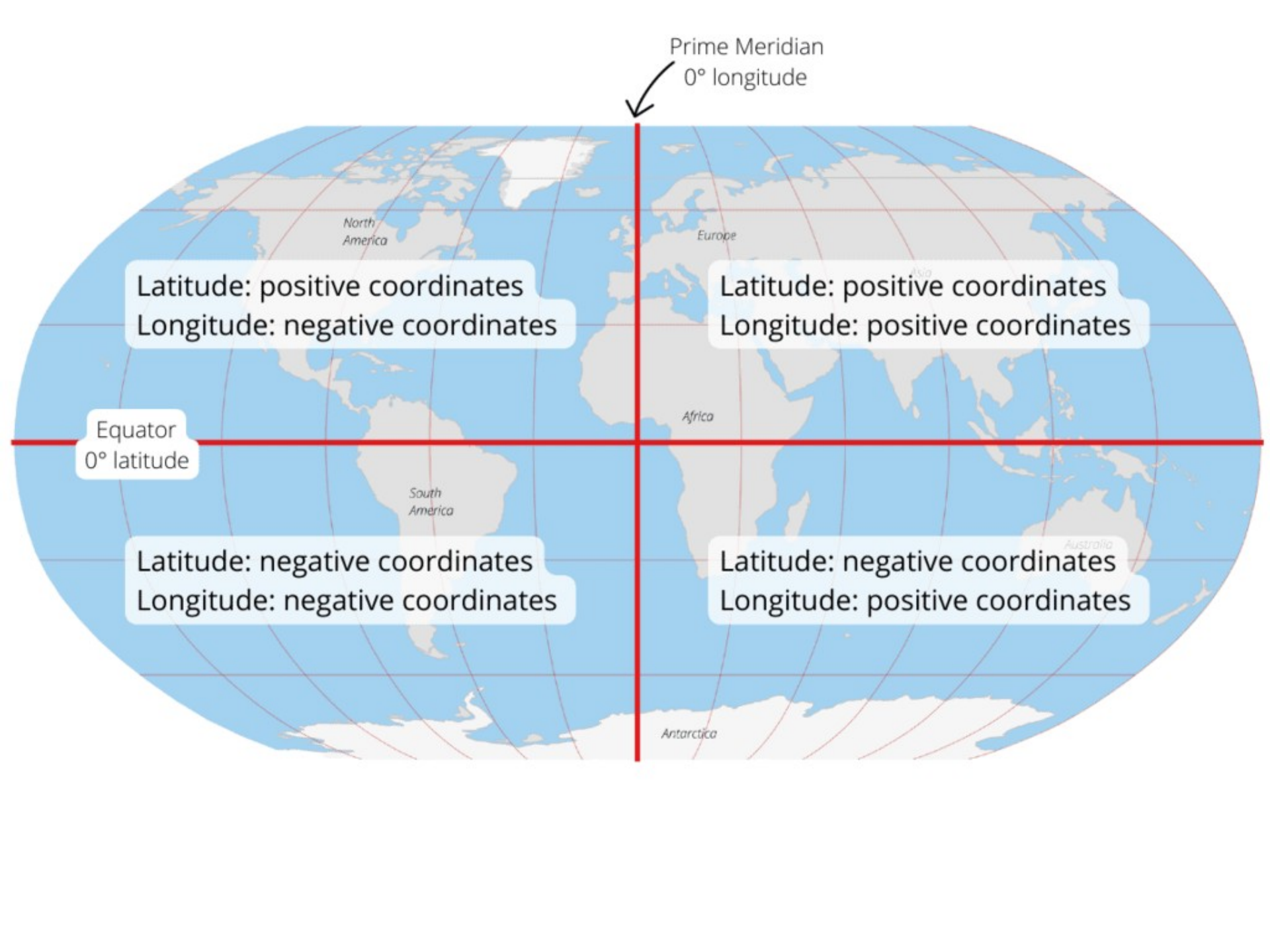
South
America

Latitude: negative coordinates
Longitude: negative coordinates

Latitude: negative coordinates
Longitude: positive coordinates

Australia

Antarctica



Degrees Minutes Seconds to Decimal Degrees

Enter Degrees Minutes Seconds latitude:			
Enter Degrees Minutes Seconds longitude:			
Convert to Decimal		Clear Values	
Results: Latitude:		Longitude:	

Decimal Degrees to Degrees Minutes Seconds

Decimal Latitude:	
Decimal Longitude:	
Convert to Degrees Minutes Seconds	Clear Values
Results: Latitude:	
Longitude:	

NADCON NAD27 - NAD83 Coordinate Conversions
from the National Geodetic Survey

<https://www.fcc.gov/media/radio/dms-decimal>

Where can you get these records?

- Field survey, observation;
- Published literatures, monographs, papers, atlases
- Museum & herbaria:



free and open access to biodiversity databases

- ▶ gbif
- ▶ iNaturalist
- ▶ VertNet
- ▶ BIEN
- ▶ ...

How to get longitude/latitude ?

- longitude/latitude --> x/y
- Online database -> longitude/latitude
- Locality -> longitude/latitude

tools to assist georeferencing information

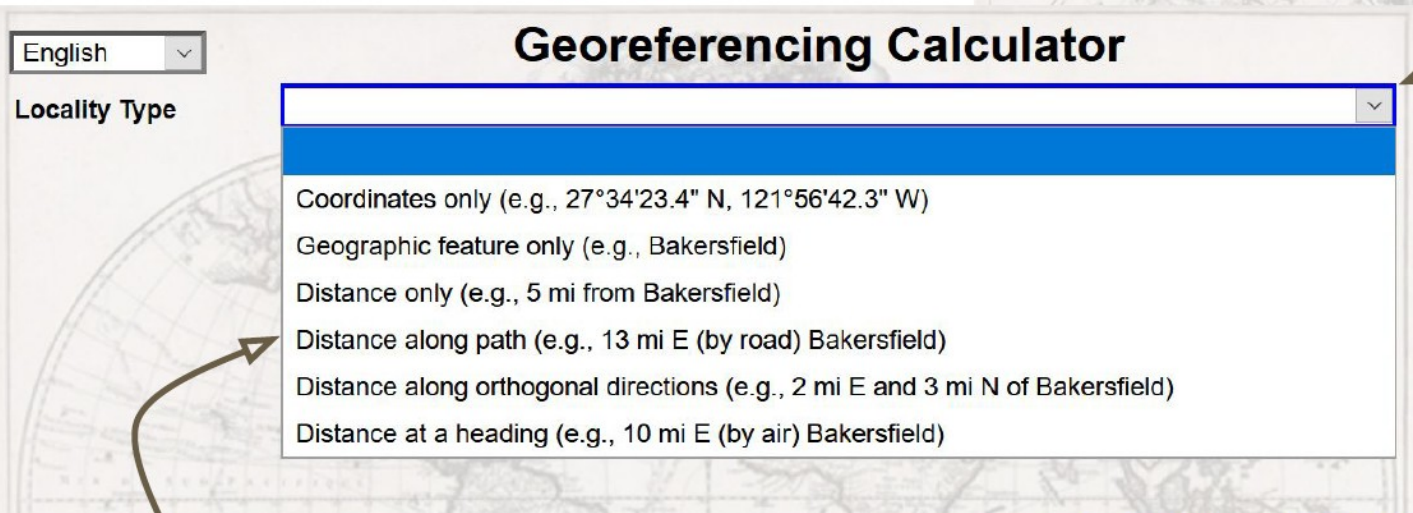
gazetteer

- ▶ BioGeomancer
- ▶ geoLoc CRIA
- ▶ GEOLocate

<http://georeferencing.org/georefcalculator/gc.html>

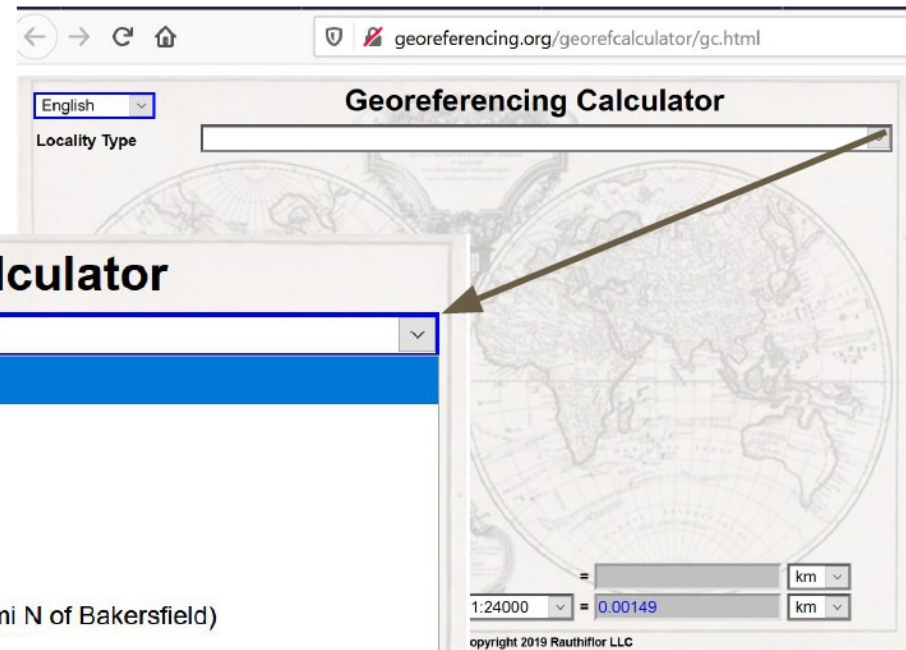
Georeferencing calculator

1. Choose locality type



The screenshot shows the 'Georeferencing Calculator' web application. On the left, there is a language dropdown set to 'English' and a 'Locality Type' label. A dropdown menu is open, displaying six options: 'Coordinates only (e.g., 27°34'23.4" N, 121°56'42.3" W)', 'Geographic feature only (e.g., Bakersfield)', 'Distance only (e.g., 5 mi from Bakersfield)', 'Distance along path (e.g., 13 mi E (by road) Bakersfield)', 'Distance along orthogonal directions (e.g., 2 mi E and 3 mi N of Bakersfield)', and 'Distance at a heading (e.g., 10 mi E (by air) Bakersfield)'. A brown arrow points from the 'Locality Type' label to this dropdown menu.

Locality description: USA, Georgia,
Augusta, Wheelless Rd., 0.5 mi S of
Gordon Hwy



This screenshot shows the main interface of the 'Georeferencing Calculator'. It includes a language dropdown set to 'English', a 'Locality Type' label, and a large text input field for the locality description. Below the input field is a world map. To the right of the map, there are two scale input fields: the first is set to '1:24000' and the second to '0.00149', both with units set to 'km'. A brown arrow points from the 'Locality Type' label to the main text input field.

[Georeferencing Guidelines \(English\)](#)

[Georeferencing Quick Reference Guide \(English\)](#)

This application was adapted by Craig Wiecezorek and John Wiecezorek from an earlier version in Java by John Wiecezorek with contributions from Qinghua Guo, Carmen Boureau, and Craig Wiecezorek. Interface translations provided by: Gustavo Iglesias (Español), Jose (Wly) Dos Santos (Português), Tendro Ramaharitra (Français), Robert Hijmans (Nederlands), Peter Desmet (Nederlands)

Preferred citation:

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Georeferencing calculator

2. Enter parameters; click Calculate

Georeferencing Calculator

English

Locality Type Distance along path (e.g., 13 mi E (by road) Bakersfield)

Coordinate Source Google Earth/Maps >2008

Coordinate Format decimal degrees

Input Latitude 33.4449

Input Longitude -82.0453

Datum (WGS84) World Geodetic System 1984

Precision 0.0001 degrees

Radial of Feature 0

Measurement Error 0.1

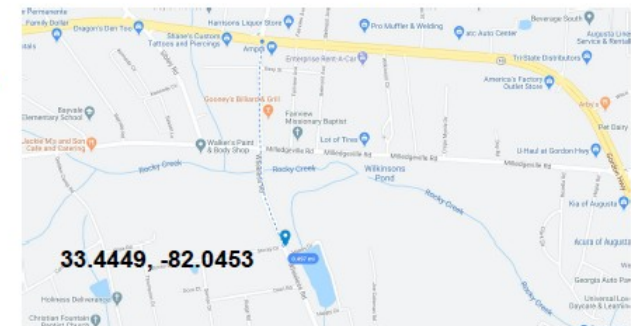
Distance Units mi

Precision 1/10 mi

Calculate Copy Go here

Locality description: USA, Georgia, Augusta,
Wheless Rd., 0.5 mi S of Gordon Hwy

Used Google Maps for coordinates



Georeferencing calculator

Results:

English

Georeferencing Calculator

Locality Type

Distance along path (e.g., 13 mi E (by road) Bakersfield)

Coordinate Source

Google Earth/Maps >2008

Coordinate Format

decimal degrees

Input Latitude

33.4449

Input Longitude

-82.0453

Datum

(WGS84) World Geodetic System 1984

Precision

0.0001 degrees

Radial of Feature

0

Measurement Error

0.1

Distance Units

mi

Precision

1/10 mi

Calculate

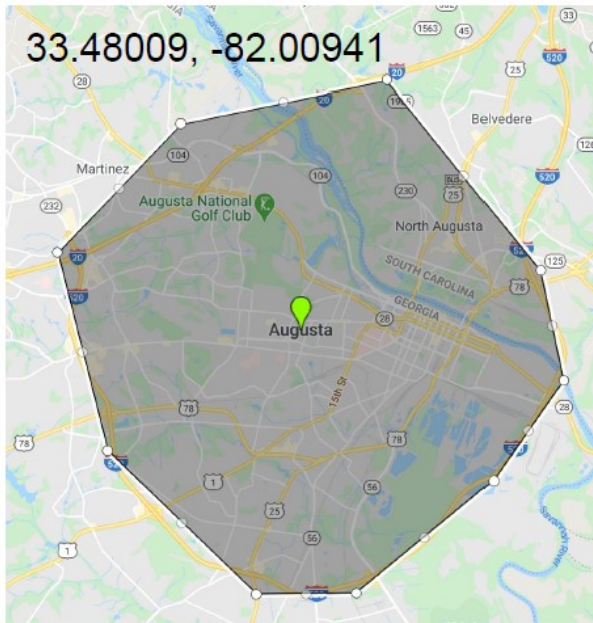
Copy

Go here

Latitude	Longitude	Uncertainty (m)	Datum
33.4449	-82.0453	264	(WGS84) World Geodetic System 1984
Precision	Date	Georeferenced by	Protocol
0.0000001	2020-03-28T03:21:11		protocol not recorded

Georeferencing calculator

What if the locality description was: USA, Georgia, Augusta, 5 mi N



Georeferencing Calculator

English

Locality Type: Distance at a heading (e.g., 10 mi E (by air) Bakersfield)

Coordinate Source: Google Earth/Maps >2008

Coordinate Format: decimal degrees

Input Latitude: 33.48009

Input Longitude: -82.00941

Datum: (WGS84) World Geodetic System 1984

Precision: 0.0001 degrees

Direction: N

Offset Distance: 5

Radial of Feature: 4.5

Measurement Error: 0

Distance Units: mi

Precision: 1/2 mi

Calculate Copy Go here

Latitude	Longitude	Uncertainty (m)	Datum
33.5526398	-82.00941	11528	(WGS84) World Geodetic System 1984
Precision	Date	Georeferenced by	Protocol
0.0000001	2020-03-28T03:46:4		protocol not recorded

Uncertainty:
11.5 km! Can't use with WorldClim 1 km

R packages for data download

Name	Description
BIEN	Tools for Accessing the Botanical Information and Ecology Network Database
biolink	Create Hyperlinks to Biological Databases and Resources
distdrawr	Download Occurrence Data of Vascular Plants in Germany from the FLORKART Database
helminthR	Access London Natural History Museum Host-Helminth Record Database
neotoma	Access to the Neotoma Paleoecological Database Through R
neotoma	Access to the Neotoma Paleoecological Database Through R
ntbox	Stable version of nichetoolbox package
occCite	Querying database aggregators and citing primary sources of resulting occurrence points
paleobioDB	Download and Process Data from the Paleobiology Database
rebird	R Client for the eBird Database of Bird Observations
rgbif	rgbif: Interface to the Global 'Biodiversity' Information Facility API
rvertnet	Search 'Vertnet', a 'Database' of Vertebrate Specimen Records
rYoutheria	Access to the YouTheria Mammal Trait Database
spocc	Interface to Species Occurrence Data Sources
vegdata	Access Vegetation Databases and Treat Taxonomy
wallace	A Modular Platform for Reproducible Modeling of Species Niches and Distributions
worrms	World Register of Marine Species (WoRMS) Client
zbank	'ZooBank' API Client

Name	Description
flora	Tools for Interacting with the Brazilian Flora 2020
rangeBuilder	Occurrence Filtering, Geographic and Taxonomic Standardization and Generation of Species Range Polygons
ritis	Integrated Taxonomic Information System Client
taxa	Taxonomic Classes
taxize	Taxonomic Information from Around the Web
taxizedb	Tools for Working with 'Taxonomic' Databases
Taxonstand	Taxonomic Standardization of Plant Species Names
vegdata	Access Vegetation Databases and Treat Taxonomy
WorldFlora	Standardize Plant Names According to World Flora Online Taxonomic Backbone
yatah	Yet Another TAXonomy Handler

Name	Description
CoordinateCleaner	Automated Cleaning of Occurrence Records from Biological Collections
KnowBR	Discriminating Well Surveyed Spatial Units from Exhaustive Biodiversity Databases
ntbox	Stable version of nichetoolbox package
rangeBuilder	Occurrence Filtering, Geographic and Taxonomic Standardization and Generation of Species Range Polygons
scrubr	Clean Biological Occurrence Records

Getting online data:

```
### Getting online data #####  
install.packages("spocc")  
library(spocc)  
  
out <- occ(query = 'Popillia japonica',  
           date = c('2020-01-01', '2022-12-31'),  
           from = c('gbif', 'bison', 'inat', 'idigbio'),  
           has_coords = TRUE, limit = 10000)  
  
dat<-occ2df(out)  
  
head(dat)  
dim(dat)  
  
write.csv(dat, "JB_2020.csv", row.names = F)
```

what the errors are

whole range of data, mostly collected opportunistically rather than systematically

museum data

- ▶ historical data w/o coordinates or locality
- ▶ inaccurate coordinates
- ▶ attributed by someone that did not collect the data

observational data

- ▶ more precise georeferencing information
- ▶ higher variance in the quality taxonomic and nomenclature information

almost nothing about absences!

spatial error

wrong or inaccurate coordinates

- ▶ misidentifying
- ▶ imprecise coordinates
- ▶ missing information inserted as zeros
- ▶ long and lat misplacement, inversion, wrong signal

taxonomic and nomenclatural error

wrong or misspelled names

- ▶ inaccurate identification
- ▶ misidentification
- ▶ synonyms and correction by specialists

Step 1: Clean errors

definition of data cleaning

determine **inaccurate**, **incomplete** or **unreasonable** data for:

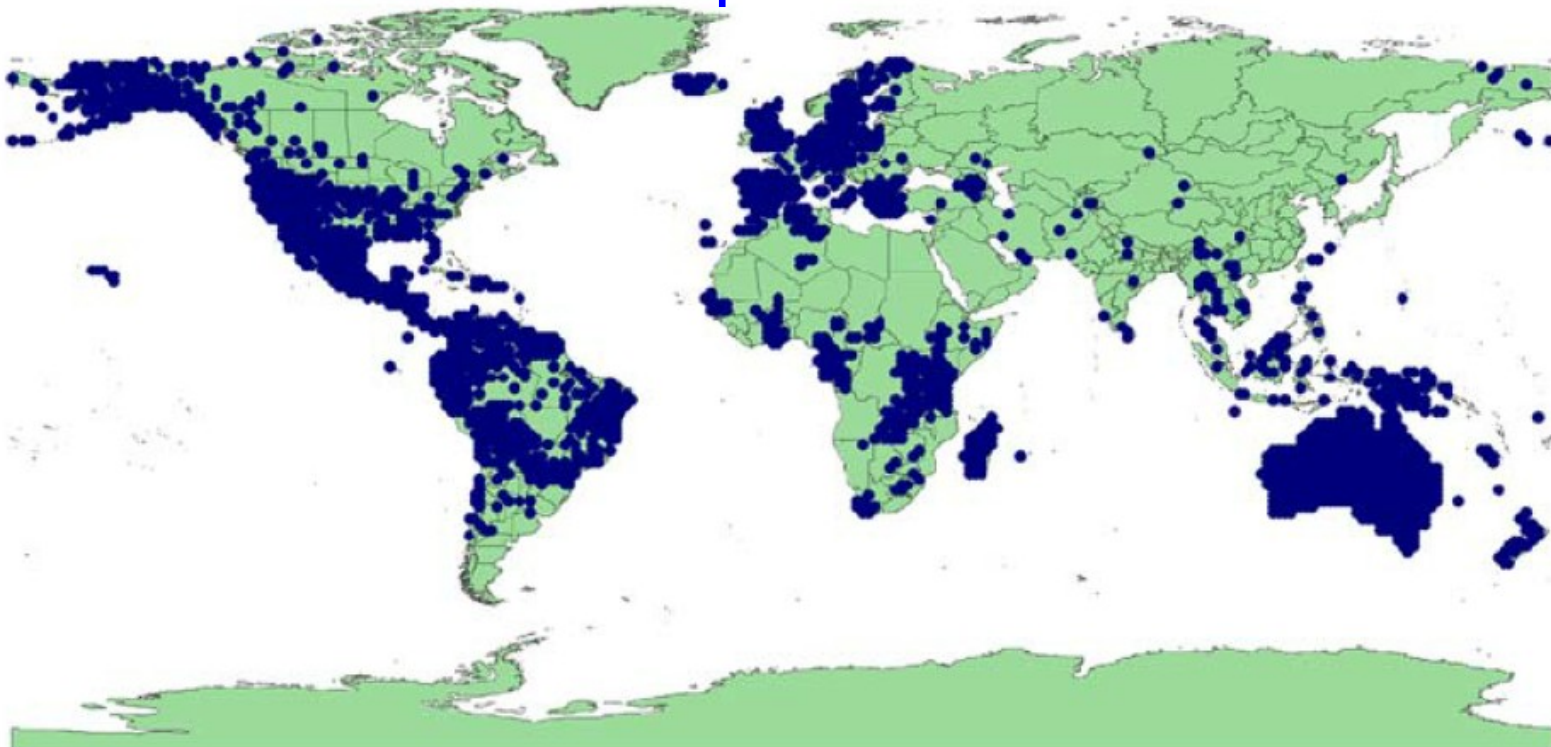
- ▶ format check
- ▶ completeness checks
- ▶ reasonableness checks
- ▶ limit checks
- ▶ outliers (geographical, temporal, environmental)



biodiversity data

Legume plant species data from gbif

Any pattern /
problem?



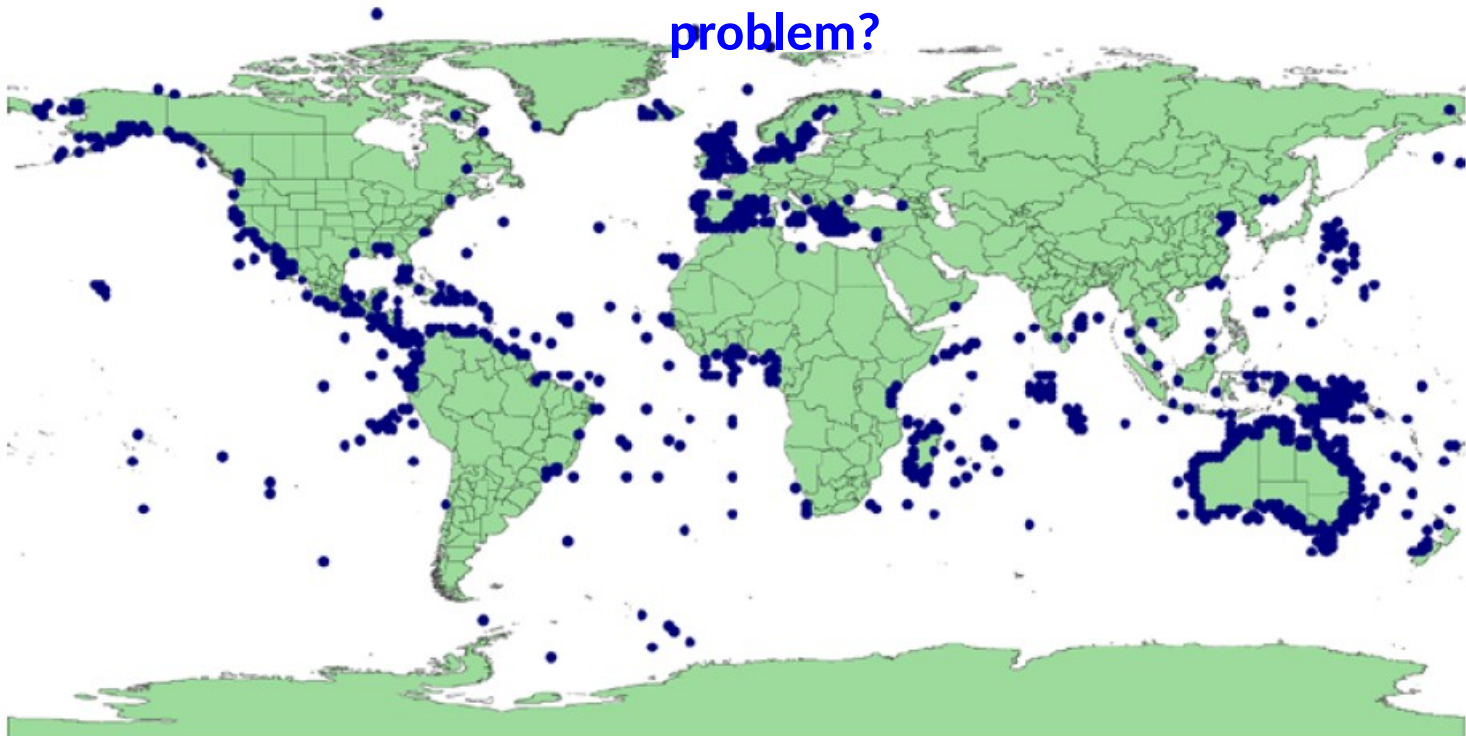
Yesson et al 2007 DOI: [0.1371/journal.pone.0001124](https://doi.org/10.1371/journal.pone.0001124)

biodiversity data

Legume plant species data from gbif



Any pattern /
problem?

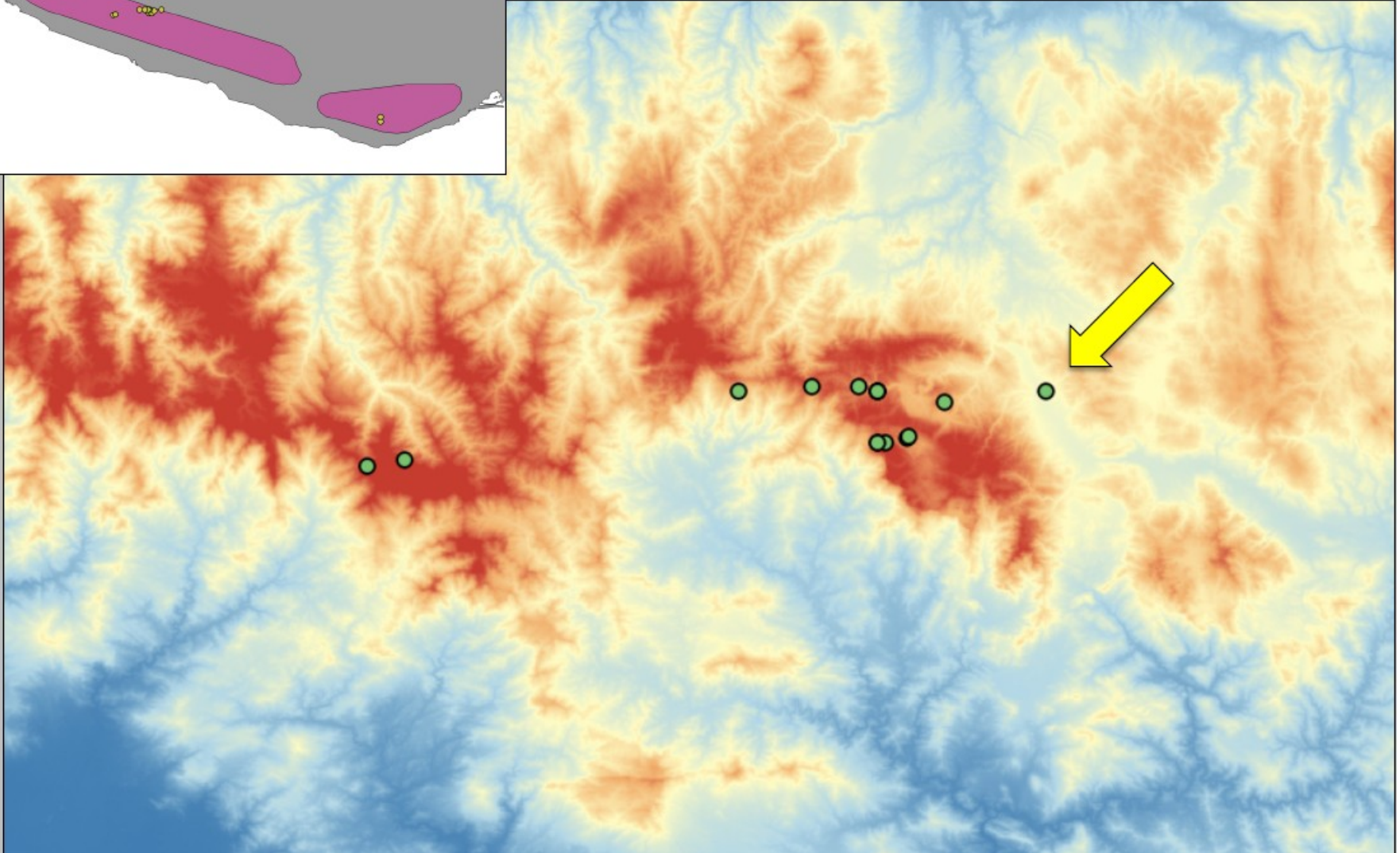
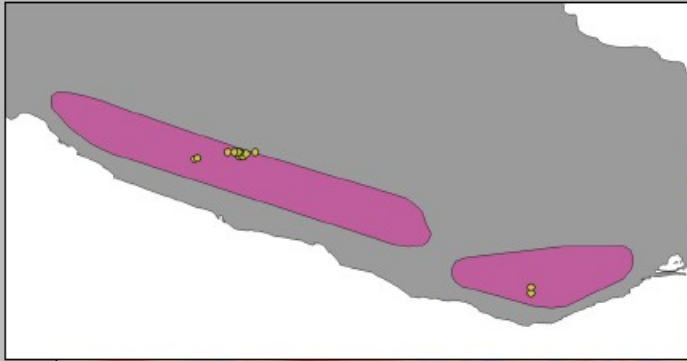


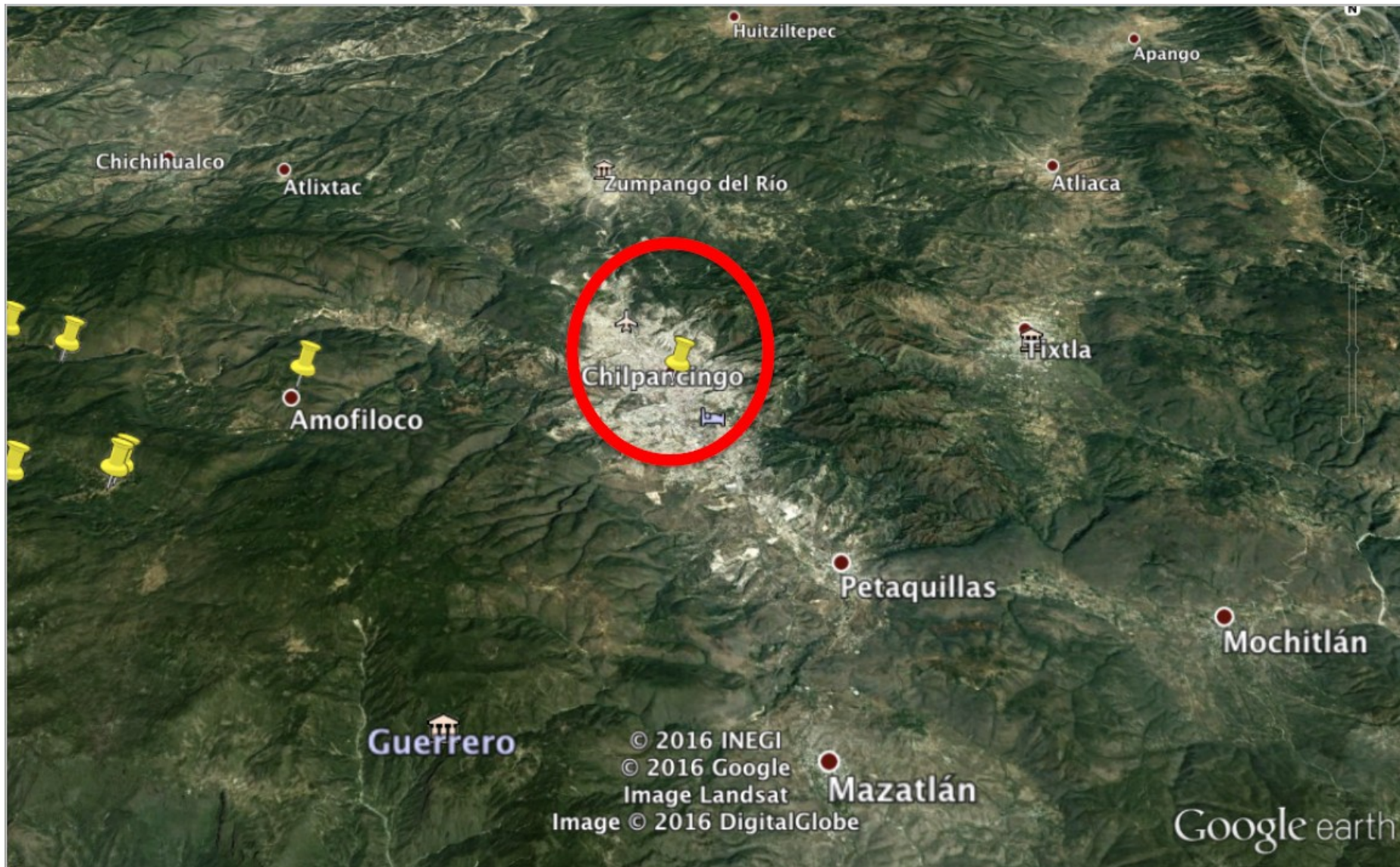
Yesson et al 2007 DOI: [0.1371/journal.pone.0001124](https://doi.org/10.1371/journal.pone.0001124)

Geographic Outliers



Environmental Outliers





Step 2: Remove bias

Occurrence filtering and dealing with bias/autocorrelation

Spatial autocorrelation

- Simplified idea - observations that are closer to each other in space may be more similar to each other
- From R. Hijmans "Spatial Data Science with R" website (<https://rspatial.org/raster/analysis/3-spauto.html>)
 - *Measures of spatial autocorrelation describe the degree to which observations (values) at spatial locations (whether they are points, areas, or raster cells), are similar to each other.*
- Spatial autocorrelation results in a violation of the assumptions of independence among observations

The First Law of Geography, according to Waldo Tobler, is "**everything is related to everything else, but near things are more related than distant things.**"

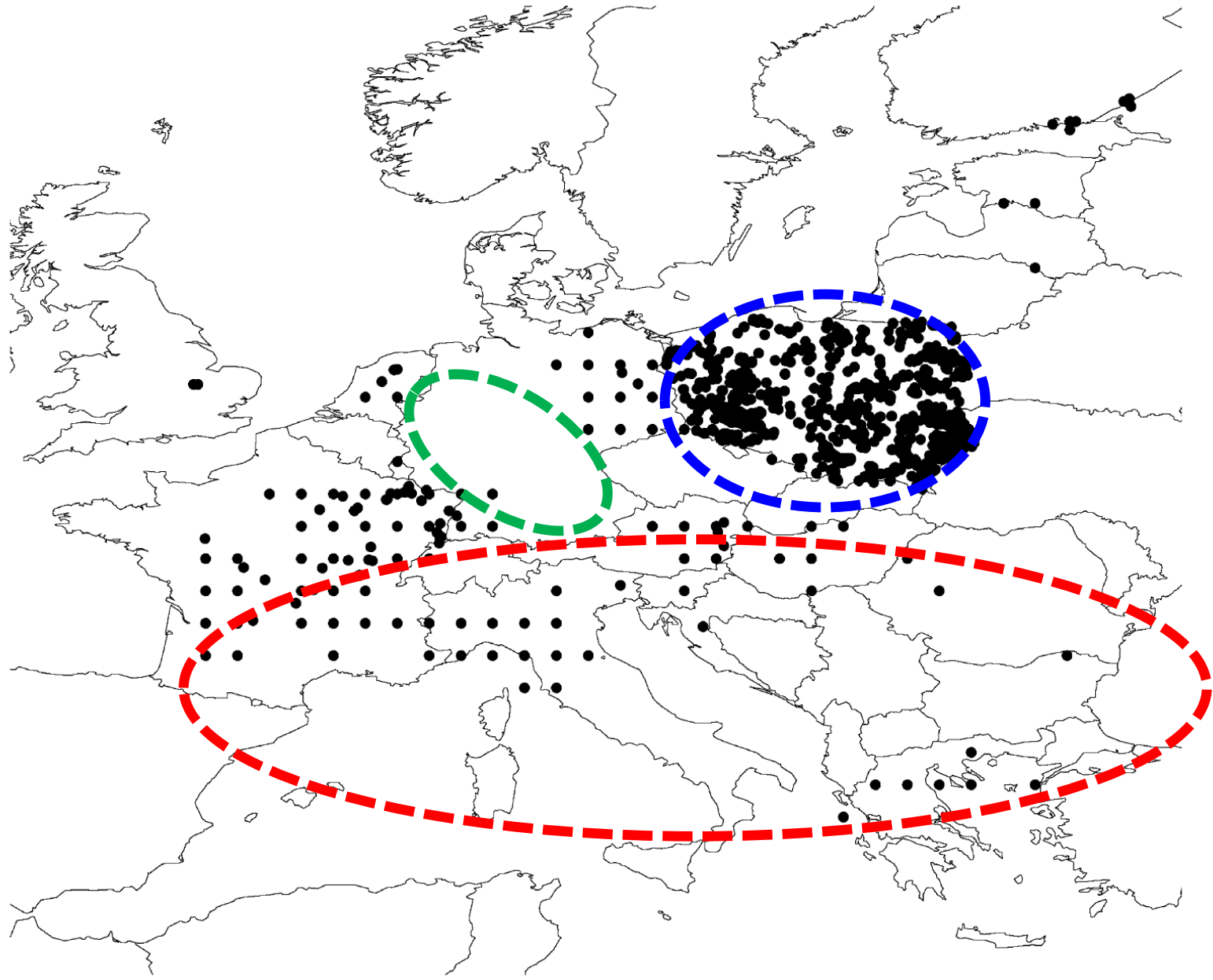


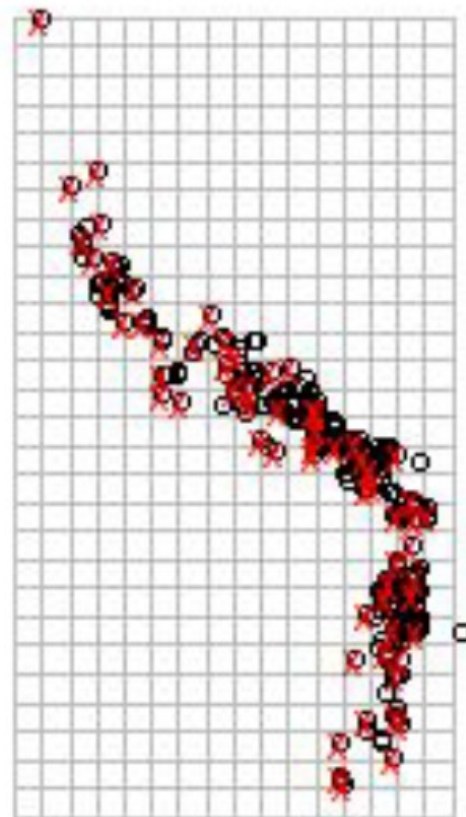
Fig. 1 Map of known occurrences of *Lycaena dispar*, based on data from the Global Biodiversity Information Facility (GBIF; <http://www.gbif.org>). The high density of points in Poland, compared with lower (or null) densities in surrounding countries illus-

trates the sampling bias. The grid-like arrangement of most points outside Poland reflecting different precision of georeferencing may compromise the results of any modeling using such data

Addressing spatial autocorrelation: Presence only models

Spatial thinning

- In geographic space
- Overlay a grid, select a set number of points from each grid cell

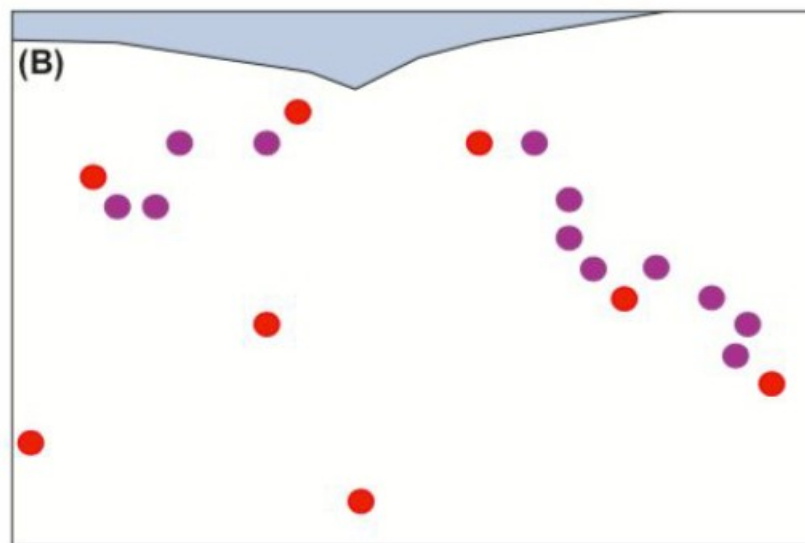


From R. Hijmans "Spatial Data Science with R"

Addressing spatial autocorrelation: Presence only models

Spatial thinning

- In geographic space
- Keep a set of points where the nearest neighbor for each point is no closer than a set distance

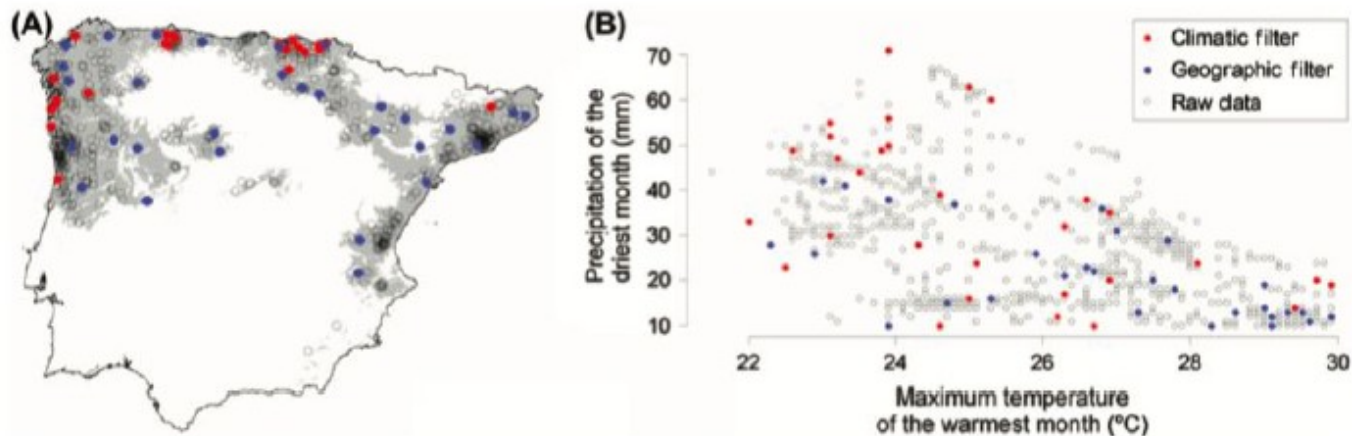


From Aiello-Lammens et al. 2015

Addressing spatial autocorrelation: Presence only models

Environmental thinning

- In environmental space
- Most *directly* addresses the effects geographic biased sampling has on model calibration

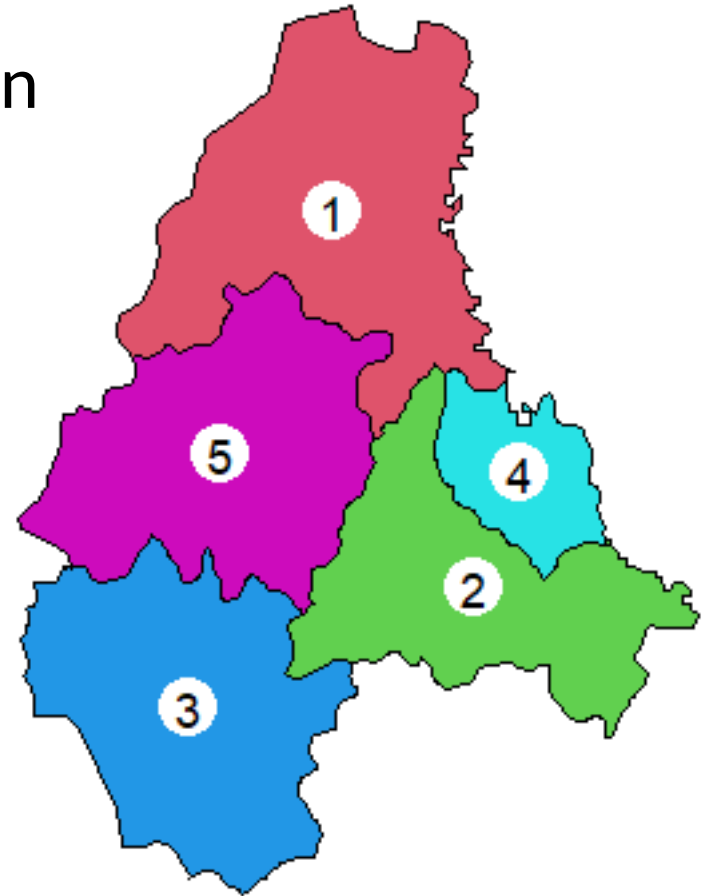


From Varela et al. 2014

Addressing spatial autocorrelation
Presence only models

Area weighted method

Select the number of points according
the areas of administration unit;
Lage state -> reserve more points;



Model calibration and evaluation strategy:

Species with many records:

e.g., invasive species, data splitting (30/70%), checkboard, block, effect of replicate (machine learning);

Species with limited records:

e.g., rare and endemic species, ensemble of small model, Pearson method (Jackknife-1)

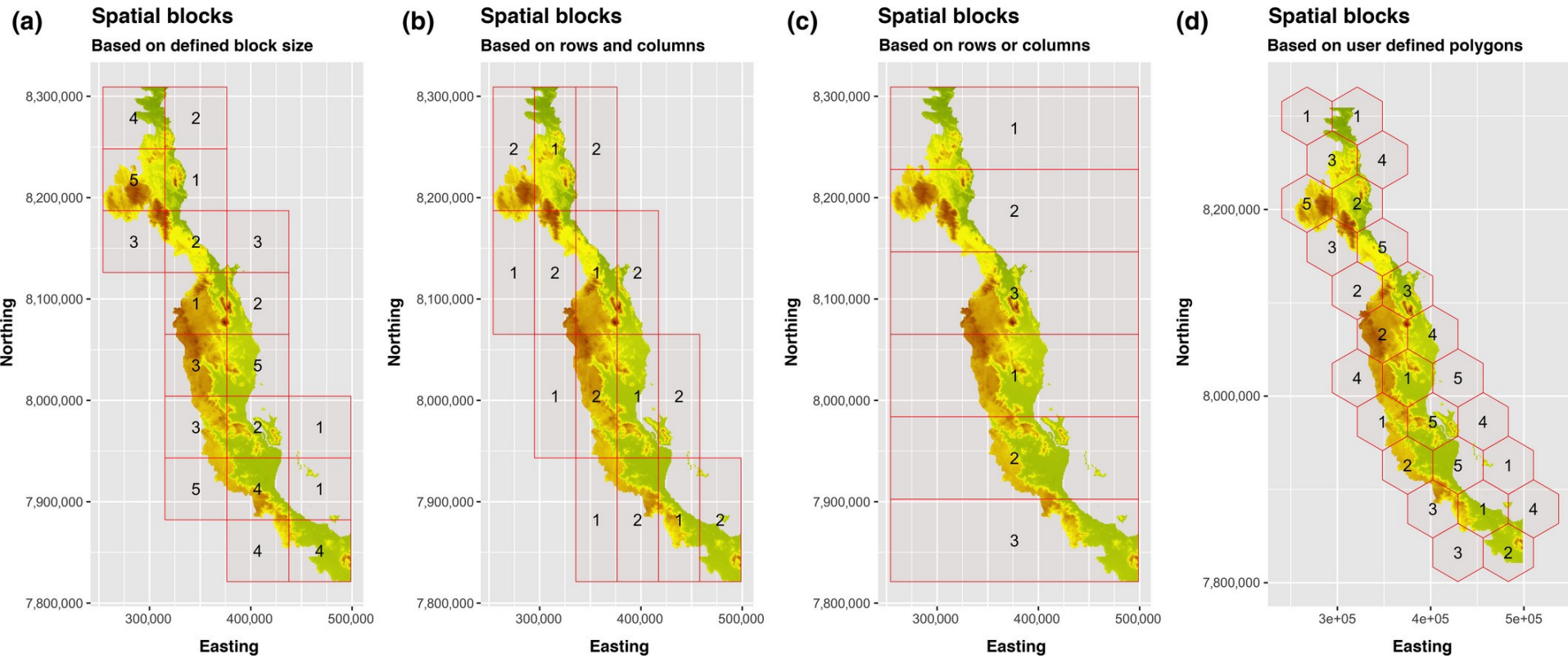
data splitting-species with many records

Model evaluation strategy

- each model should be able to accurately predict the input data (training data)
- but can each model also accurately predict new data?
- if we have independent data, we can evaluate each model on it
- if we do not, we can evaluate each model on subsets of itself
- this is called ***cross validation***

The simplest way is random data splitting (70% training, 30% testing), however they are not independent.

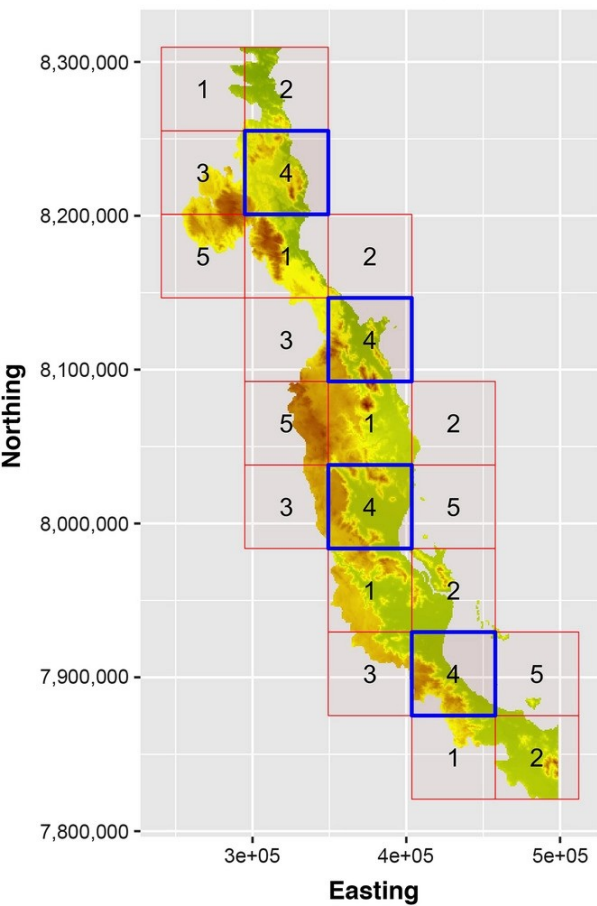
Method 1: Spatial block cross validations



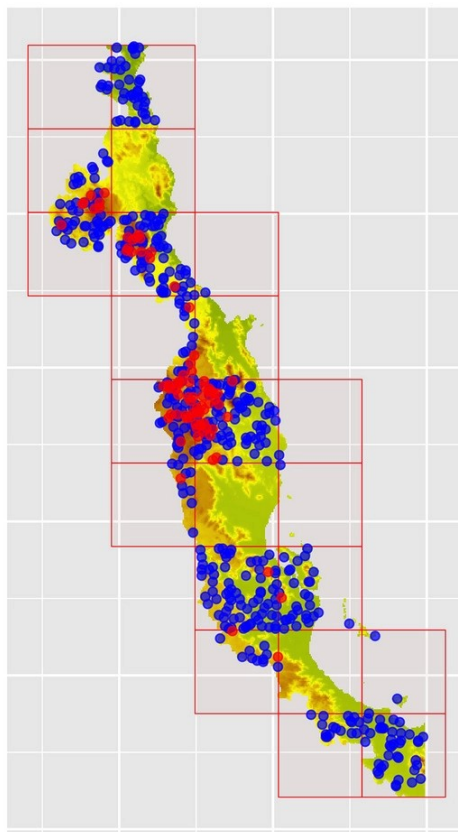
R package: blockCV

Spatial blocks

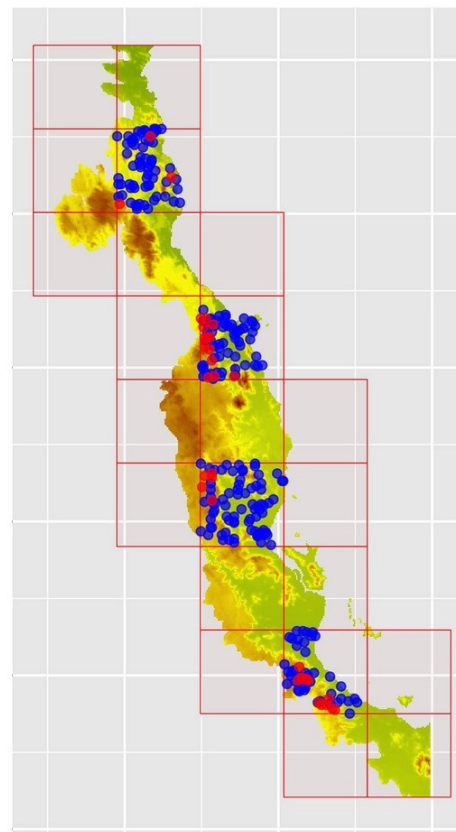
The systematic fold assignment



Training set



Testing set

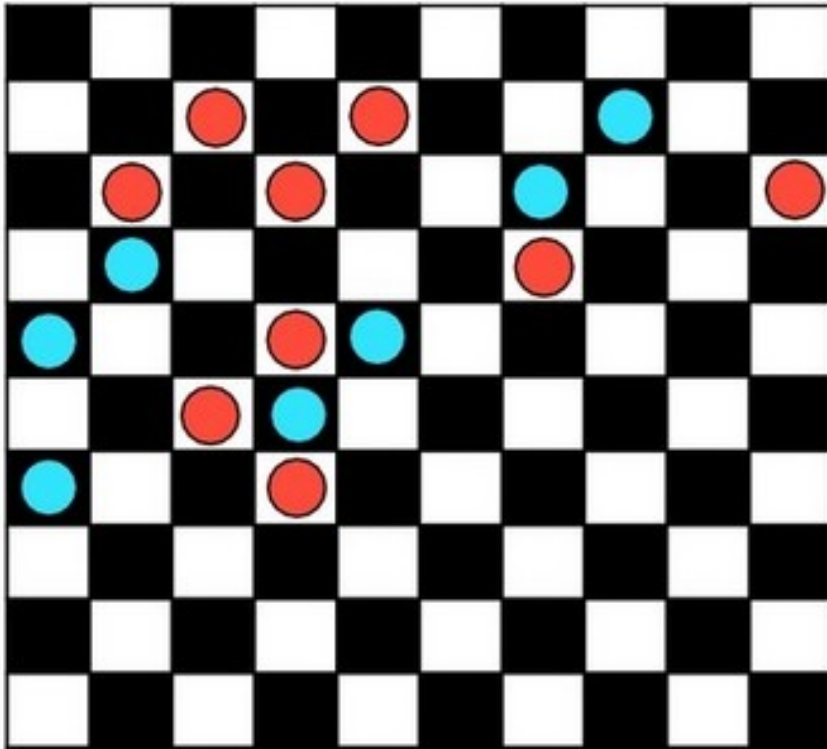


Species data

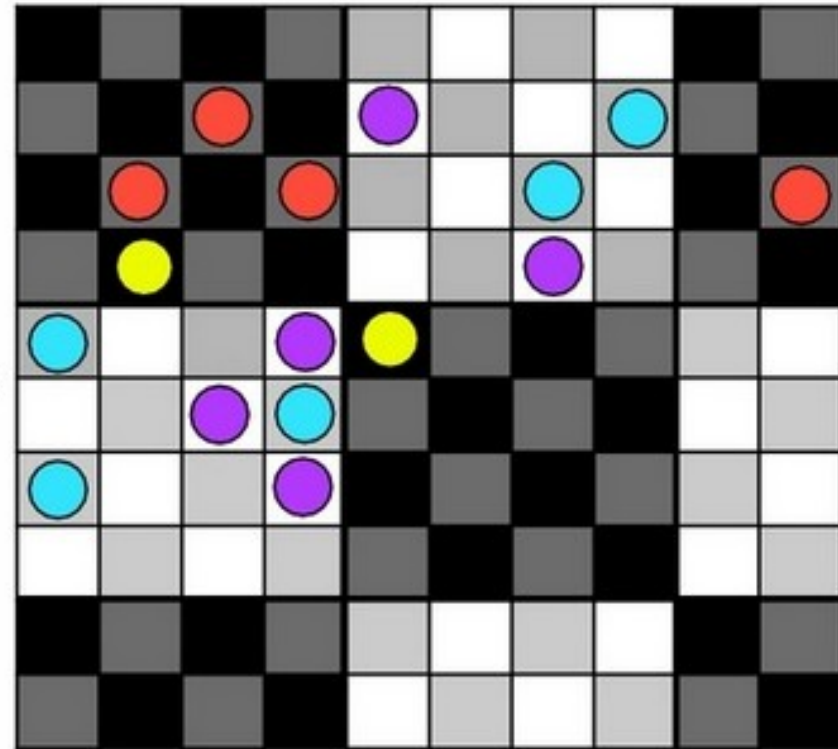
- Absence / Background
- Presence

Method 2: Spatial checkerboard

(c) 'Checkerboard1'



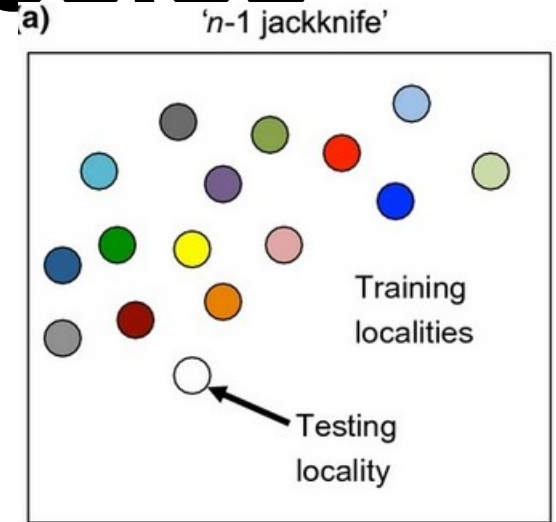
(d) 'Checkerboard2'



R package: ENMeval

Rare, endemic species

- Leaving-one-out method;
Proposed by Richard Pearson et al. 2007
lowest presence threshold (LPT)



leave-one-out (jackknife)

-
- **Ensemble of small model;**
Proposed by Guisan group
ecospat package

Building Ensembles of Small Models (ESMs)

Simple bivariate models

e.g.:

$biva_1: T_{mean} + P_{mean}$

$biva_2: T_{mean} + topo$

$biva_3: P_{mean} + topo$

etc.



Ensemble of Small Models ESM

Average of small models weighted by a score from an evaluation index

$$ESM = \frac{\sum_{i=1}^n biva_i \times EI_i}{\sum_{i=1}^n EI_i}$$

